

Write-up: next-season style intelligence (H&M data)

Deliverables: top 3 in `outputs/predictions.json` · concepts in `outputs/generated_concepts.png` · evidence in `outputs/evidence_sheet.png` and `outputs/evidence/<code>/` · API in `backend/` (API.md) · app in `frontend/app.py`. Every number below comes from a committed output file, which is named next to it.

1. Problem definition

A fashion retailer wants to know which existing styles will lead next, and wants new products that build on them.

- **Style** = `product_code` (all colourways of one design; `style_id` = 7 digits). Colourways share the design "DNA"; ranking articles would return three colours of one trouser. The winners sell in 10-23 colours, which also shows that customers buy the *shape*, not the colour. `product_code` = `article_id // 1000` for every article, so the id is stable.
- **Success** = the style's units in the next 4 weeks rank in the **top 1% of styles active at that cutoff** (sold in the previous 12 weeks). The threshold is set per cutoff (857-1,512 units across the scored cutoffs; about 194 winners of ~19,300 styles). A rank-based threshold fits the decision, because a buyer reviews a short list, and it isn't distorted by seasonal swings or the 2020 COVID dip.
- **Prediction period** = the 4 weeks after the cutoff. Data ends 22 Sep 2020, so the forecast covers **23 Sep - 20 Oct 2020**, the opening of autumn. Styles turn over fast: the median style sells for 19 active weeks, and 32% of weekly units come from styles launched in the last 12 weeks. Over a 13-week season much of the demand comes from styles that don't exist yet, so 4 weeks is the longest window where "which existing styles will lead" is still well defined and testable. "Next season" is carried by the concepts, which are new products built on the winners' proven DNA.

2. Data & EDA

Used in full: all 31.8M transactions (20 Sep 2018 - 22 Sep 2020), 105,542 articles and 1.37M customers, kept outside the repo and converted to Parquet. **Images:** only the 9 reference photos of the 3 winners, plus one list photo per top-50 style for the app (the full set is about 30 GB). The forecast doesn't use images, so the ranking is unaffected. The concepts only see each winner's best-selling colours.

Findings (`outputs/figures/eda_insights.md` , `data_science/notebooks/02_eda_extended.ipynb`):

- **Seasonality:** weekly units swing 3.6× (177k to 636k). Swimwear is 38% of the predicted summer top-100 and 0% in autumn.
- **Long tail:** in the last 52 weeks the top 1% of styles took 31% of units.
- **Newness:** 32% of weekly units come from styles under 12 weeks old (range 20-46%).
- **Customers:** age is bimodal (peaks at 21 and 51; 1.16% missing). Ages 25-34 buy 36.4% of units. Online is 66.7-74.8% of units in every age band (69% → 73% from 2019 to 2020).
- **Data quality:**
 - *Duplicate rows:* 9.36% of transaction rows exactly repeat another row (same customer, day, article, price, channel). The raw CSV gives the same count, they occur on every day, and their frequency falls steeply with group size, which fits multiple units in one purchase. Each row is kept as one unit.

- *Prices* are scaled by Kaggle (max ≈ 0.59). Within product type, 0.26% are low outliers and 0.04% high; they are kept.
- *Coverage*: no missing dates. 995 articles never sold, and 0.39% of articles lack a description.

3. Approach & model selection

Transactions are aggregated with DuckDB to a weekly style table (Wed → Tue weeks). For each weekly cutoff, a snapshot holds **23 features computed only from weeks before the cutoff**:

- momentum: units over 1/2/4/8/12 weeks, plus trend ratios;
- buyers, repeat rate and online share;
- price and discount vs peak price;
- weeks since launch and same period last year;
- six product attributes.

The target is units in the 4 weeks from the cutoff. There are 45 training snapshots (2019-09-25 → 2020-07-29), and tests scramble future data to check that the features don't change.

- **Regressor for ranking.** LightGBM predicts $\log_{1p}(\text{next 4 weeks}) - \log_{1p}(4 \times \text{last week})$, the uplift over the naive run-rate, weighted toward high-volume styles. A plain log-units regressor lost to "last week $\times 4$ " at the top of the ranking, and a LambdaRank variant lost on the inner cutoffs (NDCG@50 0.846 vs 0.944). The regressor gives unit forecasts, which a merchandiser plans with, and it ranks the list.
- **Classifiers for calibrated scores.** Two LightGBM binary models (top 1% and top 0.1% labels) are trained on the same features and folds. Their probabilities are calibrated with isotonic maps fitted **only on out-of-fold predictions whose labels were known at each cutoff** (OOF cutoff $\leq$ cutoff – 4 weeks).
 - P(top 1%) saturates at 1.0 for 17 of the regressor's top 20 styles.
 - P(top 0.1%) spreads out (10 distinct values, 0.038–1.000), so it is the displayed `prediction_score`.
 - P(top 1%) is kept as `confidence_top1pct`.
- **Selection:** highest forecast units, at most one style per garment group, sold in the last 2 weeks (the only availability signal).

4. Results

Every model is retrained at each cutoff on cutoffs whose target ended before it: a 10-cutoff rolling backtest (2020-05-27 → 2020-07-29) plus the validation week (cutoff 2020-08-26).

Backtest mean (10 cutoffs)	Model	Last week $\times 4$	Last 4 weeks	Regressor
Regressor NDCG@50	0.924 $\pm$ 0.031	0.906 $\pm$ 0.040	0.906 $\pm$ 0.016	–
Regressor precision@12	0.725	0.708	0.658	–
Classifier top 1%, PR-AUC	0.770	0.738	0.713	0.771
Classifier top 0.1%, PR-AUC	0.807	0.750	0.743	0.796

Sources: `outputs/figures/eval_table.md`, `outputs/classifier/eval_classifier.md`, `eval_classifier_top0.1pct.md`.

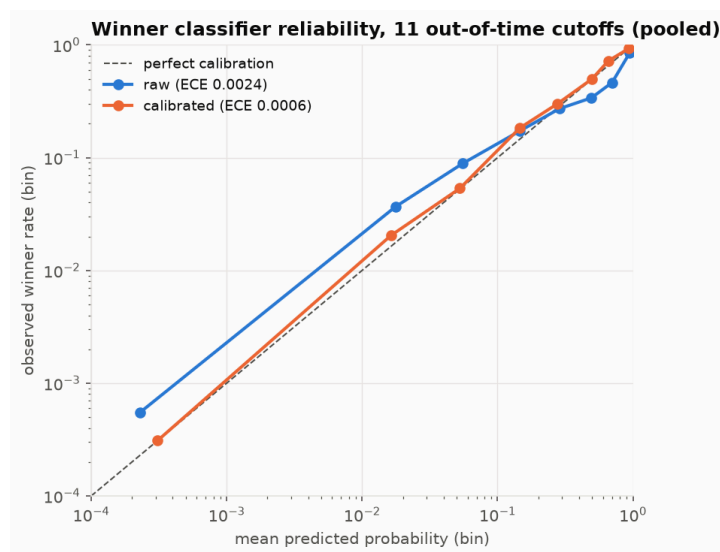
- **Regressor vs naive baselines:**
 - It wins 7/10 cutoffs on NDCG@50 against both baselines.
 - On precision@12 it beats last week $\times 4$ at only 3/10 (6 ties).

- On the validation week it **ties last week × 4** (NDCG@50 0.862 vs 0.864) and clearly beats last 4 weeks (0.569).
- Its value is **demoting fading styles**: where it disagrees with last week × 4, its demotions were right 6/10, its promotions 3/10 (`outputs/figures/movers.md`).

- **Classifiers:**

- They beat the naive baselines: top 1% is ahead of last week × 4 on PR-AUC at 9/10 cutoffs.
- They **tie the regressor** (0.770 vs 0.771), so they don't rank better.
- The top-0.1% model even loses to last week × 4 on the validation week (PR-AUC 0.456 vs 0.566). Its score is a relative-strength signal, not a better ranker.

- **Calibration:** isotonic lowers the top-1% ECE from 0.00361 to 0.00282 (backtest) and from 0.00271 to 0.00225 (validation). For top 0.1% the change (0.00041 → 0.00044) is within noise (2 standard errors = 0.00006), and Brier improves at 8/11 cutoffs.



Seasonal check (bonus): the same pipeline at the 27 May 2020 cutoff put 7 of its predicted top 10 in the actual top 10 (`outputs/figures/seasonal_comparison.md`). A low prediction score next to a large unit forecast was a warning sign: of the 20 largest SS2020 forecasts, the 6 scored below 0.3 include all 4 that sold under half their forecast (all C Lolly swimwear, e.g. C Lolly Top: 8,901 forecast, 1,811 sold, score 0.09), while the other 14 sold at least 84% of forecast. This is one season and 4 cases, and lower in the list low scores also cover styles that sold 2–3× their forecast (Kelso, Therese tee), so it is a flag to check stock and trend, not a correction.

5. Top 3 and why

#	Style	Forecast (last 4 wks)	prediction_score	Main SHAP drivers (× on last week × 4)
1	Pluto RW slacks, Trousers	7,417 (9,182)	1.00	last week 1,711 × 0.62 · momentum 0.75 × 0.74 · discount 2% × 1.21 · last 2 weeks × 1.16
2	Lucy blouse, Blouses	6,492 (6,662)	0.94	last week 1,705 × 0.62 · momentum 1.02 × 0.64 · discount 1% × 1.23 · last 2 weeks × 1.15
3	RICHIE HOOD, Jersey Basic	5,486 (6,108)	1.00	last week 1,165 × 0.62 · momentum 0.76 × 0.75 · discount 1% × 1.24 · last 2 weeks × 1.16

- **What they share:** all three are early-autumn risers (last 4 weeks = 1.9–2.6× the 4 weeks before) whose September rise the model partly discounts. They win on sustained volume at

near full price (1–2% below peak), not markdowns.

- **Why perennial basics aren't there:**

- Jade HW Skinny Denim (forecast rank 2) is a trouser, and Pluto has the higher forecast (7,417 vs 6,854).
- Cat Tee is in RICHIE's garment group with far fewer forecast units (2,682, rank 29).
- Long-running basics also teach the design team little that's new. That is a judgement, not a rule: RICHIE has sold for 97 weeks too.

Per-style drivers, history and lineage: `outputs/evidence/forecast.json` and the app's Style detail page.

6. Concepts and how each links to its prediction

Next-Season Concepts: Autumn 2020

Top-3 forecast winners for 23 Sep–20 Oct 2020 (4-week horizon) → AI-generated next-season concepts

#1 Pluto RW slacks · Trousers style 0751471	#2 Lucy blouse · Shirt style 0762846	#3 RICHIE HOOD · Hoodie style 0685814	
 	 	 	
WINNING STYLE	NEW CONCEPT Critic: approved	WINNING STYLE	NEW CONCEPT Critic: not approved
Why it won Rank #1 of 20,318 styles scored (one per garment group): 7,417 units forecast for 23 Sep–20 Oct 2020 (9,182 in the last 4 weeks). Model: +8% vs the last-week × 4 run-rate of 6,844; top upward driver: still near full price, only 2% below its peak price (×1.21).	Why it won Rank #2 of 20,318 styles scored (one per garment group): 6,492 units forecast for 23 Sep–20 Oct 2020 (6,662 in the last 4 weeks). Model: -5% vs the last-week × 4 run-rate of 6,820; top upward driver: still near full price, only 1% below its peak price (×1.23).	Why it won Rank #3 of 20,318 styles scored (one per garment group): 5,486 units forecast for 23 Sep–20 Oct 2020 (6,108 in the last 4 weeks). Model: +18% vs the last-week × 4 run-rate of 4,660; top upward driver: still near full price, only 1% below its peak price (×1.24).	
What changed Charcoal-and-camel glen-check wool-look fabric; camel contrast side stripe down each leg. Kept: slim tapered ankle-length leg, belt-loop waistband, slant pockets and pressed centre crease.	What changed Deep burgundy satin with a soft sheen; fuller sleeves gathered into deep buttoned cuffs. Kept: shirt collar, open V-neck, button front and rounded hem.	What changed Heather-grey fabric; twin zip pockets on the pouch; contrast-tipped ribbed cuffs. Kept: drawstring hood and kangaroo pouch. Intended cropped, boxy shape was not achieved by the image model; critic did not approve. The small chest label is still visible.	

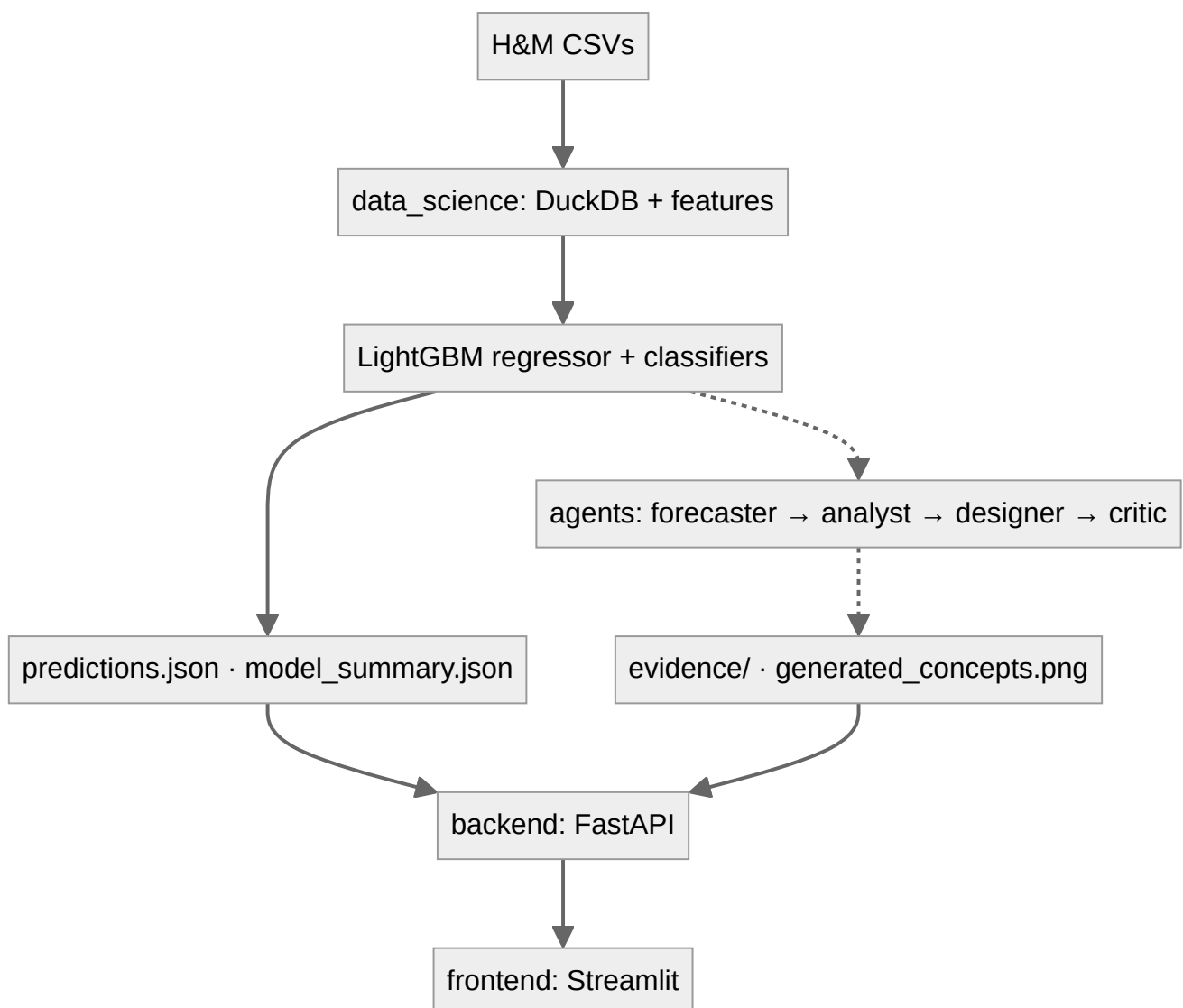
How each concept is built. For each winner, the brief (`brief.json`) KEEPs what the data says sells and CHANGEs fabric, colour, trims or proportion.

- **KEEP:** the shape that sells across colourways, the details visible in the photos and description, and the near-full-price quality.
- **Image:** FLUX.1 Kontext edits the best-selling colourway's photo.
- **Critic:** checks novelty with CLIP similarity to the style's own photos, plus a visual check.
- **Lineage:** `lineage.json` links forecast → brief → prompt → critic → board caption, and lists briefed changes the image didn't render.

Winner	Why it won (forecast)	Concept	Critic	CLIP to own photos
Pluto RW slacks	7,417 units; near full price	glen check, camel side stripe	approved	0.726
Lucy blouse	6,492 units; near full price	burgundy satin, fuller gathered sleeves	approved	0.690
RICHIE HOOD	5,486 units; near full price	grey, zip pockets, striped cuffs	not approved	0.797

RICHIE: the brief asked for a cropped, boxy shape, which the image model never produced in three attempts, so the board says the critic did not approve it. **Pluto:** it was regenerated from the run's own brief so that its lineage is clean. Both interventions were approved by the user and logged.

7. Architecture



- **data_science/:** data layer, features, regressor with backtest (`model.py`), classifiers and calibration (`classify.py`). `train.py` runs the pipeline; `predict.py` and `summary.py` write the JSON files the API serves. Nothing in the API retrains.
- **backend/:** `model_service.py` loads the predictions once and has no web code; `api.py` handles HTTP only (validation, 404/422 JSON, CORS, images); `schemas.py` holds the Pydantic models.

- Endpoints: `/styles/top`, `/styles/{id}`, `/seasons`, `/model/summary`, `/health`, `/images`. Display names drop the copy number (`raw_name` keeps it); both `/styles` endpoints take `?season=AW2020|SS2020`.
- **frontend/:** Streamlit, talking only to the API; images are fetched server-side. Light and dark theme, a sidebar glossary and tooltips on every metric.
 - Pages: overview (KPIs, the 3 picks, how to read the scores), top styles (search, filters, sort, CSV download), style detail (chart, reasons, concept, previous/next; actuals for SS2020), model performance (tiles, charts vs baselines, exact tables), seasonal view, concepts.
- **Agentic layer (bonus):** a Claude Agent SDK orchestrator with 4 sub-agents, 3 MCP servers (`retail`, `forecast`, `image`) and the `style-dna-brief` skill.
 - Guard-rails are in code: a permission gate (writes only under `outputs/`), an image budget (≤ 4 new images, ≤ 1 revision), and lineage built from files.
 - The run took 7 minutes: 59 tool calls and 3 GPU generations.

8. Limitations

- **No stock data → censored demand.** A sold-out style looks like a weak seller, so the model learns supply-limited styles as decliners. "Sold in the last 2 weeks" only removes styles that are clearly gone; no inventory assumptions were added. **Data that would help:**
 - stock and availability by store/size;
 - the markdown calendar and promotions;
 - returns;
 - web traffic (views, add-to-cart, searches);
 - store footfall;
 - size curves.
- **Horizon:** validated for 4 weeks only. A season-long forecast was not tested, so no accuracy claim is made for it.
- **Cold start:** new styles have no history, and 32% of units come from them. Only existing styles can be ranked.
- **Small gains over the naive baseline:** last week $\times 4$ is strong, and 2020 is COVID-distorted.
- **Images:** FLUX Kontext changes fabric, colour and trims, but didn't change the hoodie's proportions, and small chest labels reappear. CLIP barely reacts to colour (colourways of one style score 0.80–0.94), so the critic also checks images visually. The brief validator doesn't catch KEEP-vs-CHANGE contradictions (Pluto KEEPs "solid colour" while changing to a check).

9. Future improvements

- **Stock-aware selection:** plug in inventory and open-to-buy, and model demand as censored.
- **Cold start:** attribute-similarity to past launches, so new styles can be scored.
- **Longer horizons:** a season-level forecast with its own backtest.
- **Concepts:** a shape check in the critic (silhouette/edge comparison) instead of relying on CLIP.
- **Workflow:** route concepts to a buyer approval queue with the lineage attached.
- **Monitoring:** retrain weekly and track calibration drift.